🌡️ Temperature Converter

A modern, responsive, and user-friendly **Temperature Converter** built using **HTML5, CSS3, and JavaScript**. This web application allows users to instantly convert temperatures between **Celsius (°C)**, **Fahrenheit (°F)**, and **Kelvin (K)** through an intuitive interface.

📖 Project Overview

The Temperature Converter is a beginner-friendly web application that demonstrates JavaScript fundamentals such as user input validation, conditional logic, DOM manipulation, and event handling. It provides accurate temperature conversions with a clean, responsive design that works seamlessly across desktop and mobile devices.

This project was developed as part of my web development internship to strengthen front-end development skills.

✨ Features

- 🌡️ Convert Celsius to Fahrenheit and Kelvin
- ❄️ Convert Fahrenheit to Celsius and Kelvin
- 🔥 Convert Kelvin to Celsius and Fahrenheit
- ✔️ Input validation for numeric values
- ⚡ Instant temperature conversion
- 📱 Fully responsive design
- 🎨 Clean and modern user interface
- 🚫 Displays error messages for invalid input

🛠️ Technologies Used

- HTML5
- CSS3
- JavaScript (ES6)

📁 Project Structure

```
Temperature-Converter/
│
├── index.html      # Main webpage
├── style.css       # Styling and responsive design
├── script.js       # Temperature conversion logic
└── README.md       # Project documentation
```

🚀 Getting Started

Clone the Repository

git clone https://github.com/your-username/temperature-converter.git

Run the Project

1. Open the project folder.
2. Double-click **index.html** or open it in your preferred web browser.
3. Enter a temperature value.
4. Select the input and output temperature units.
5. Click the **Convert** button to view the converted result.

No additional software or dependencies are required.

📱 How It Works

1. Enter a temperature value in the input field.
2. Select the unit you are converting **from**.
3. Select the unit you want to convert **to**.
4. Click the **Convert** button.
5. The converted temperature is displayed instantly with the correct unit.
6. If an invalid value is entered, an error message prompts the user to enter a valid number.

📚 Learning Outcomes

This project helped in understanding:

- Semantic HTML structure
- CSS styling and responsive layouts
- JavaScript functions
- DOM manipulation
- Form validation
- Conditional statements (if...else)
- Mathematical calculations
- Event handling
- User-friendly interface design

🔮 Future Enhancements

- Real-time conversion while typing
- Conversion history
- Copy result to clipboard
- Dark/Light mode toggle
- Temperature scale information
- Animated transitions
- Voice input support
- Keyboard shortcuts

📸 Screenshot

Add screenshots of your application here.

screenshots/

└── temperature-converter.png

🎯 Project Purpose

The purpose of this project is to demonstrate the implementation of temperature conversion logic using JavaScript while building a responsive and interactive web application. It showcases essential front-end development concepts, including input validation, DOM manipulation, and responsive web design.

👨‍💻 Author

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🤝 Contributing

Contributions, suggestions, and improvements are welcome. Feel free to fork this repository, make enhancements, and submit a pull request.